



Amrita School of Artificial Intelligence

COIMBATORE CAMPUS

UNIVERSITY OF CALICUT · 24 SEPTEMBER 2026

AI in Material Science

The afternoon: what machine learning is doing in chemistry and materials right now, and two sessions where you build it yourself.

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Amrita School of AI · University of Calicut, 24 September 2026

WHERE WE ARE GOING

The afternoon

14:00 AI and multi-omics **PROF. SAI SUNDARAKRISHNA**

14:30 What AI does in chemistry and materials today

15:00 Hands-on 1, molecules in Python

15:50 Tea

16:00 Agentic AI in chemistry

16:15 Hands-on 2, an agent that does chemistry

16:50 Where this is going

This morning you learned to solve $Ax = b$.

This afternoon the rows of A are molecules.

PART 1

What AI does in chemistry today

Making a molecule is slow. Predicting it is fast.

A new material takes **10 to 20 years** from idea to product. Most of that is synthesis and testing, one candidate at a time.

The chemical space of small drug-like molecules is about **10^{60}** molecules. Every compound ever made by humans is roughly **10^8** .

We have searched almost none of it.

The entire field is one idea: **learn the structure-to-property map from the experiments already done, then search in software instead of glassware.**

STEP ONE, AND THE ONE THAT MATTERS MOST

A molecule has to become a vector

Representations

SMILES, the string

```
CC(=O)Oc1ccccc1C(=O)O
```

Compact, human-readable, and what almost every dataset ships. A bad edit gives an invalid molecule.

SELFIES, the robust string

Every possible string is a valid molecule. This is why generative models prefer it.

Descriptors, the summary

Molecular weight, logP, polar surface area, ring count. A handful of numbers a chemist already understands.

Fingerprints and graphs, the structure

Which fragments are present, or atoms as nodes and bonds as edges. This is what modern models use.

The representation decides the ceiling.

The model choice decides how close you get to it.

Property and reaction prediction

Give the model a structure, ask for a number or a label.

- **Solubility, toxicity, melting point, band gap.** You will fit the first of these yourself in forty minutes.
- **Chemprop** (MIT): a message-passing network over the molecular graph. Trained on 2,335 molecules, it picked out **halicin** from a library of 6,111, an antibiotic with a structure no chemist would have proposed. A later screen of 107 million molecules gave eight more candidates.
- **Reaction outcome and yield**, and which atom of a molecule reacts. Chemists call that selectivity, and it decides whether a synthesis works.

MATURE. IN DAILY INDUSTRIAL USE.

Machine-learned interatomic potentials

To know how atoms move you need the energy and the forces. Quantum mechanics gives them accurately and **far too slowly**: hundreds of atoms, picoseconds of time.

A neural network trained on quantum calculations reproduces those energies at a **millionth of the cost**.

- **MACE, NequIP, Allegro**: equivariant graph networks, now standard.
- **Foundation models for atoms**: MACE-MP-0, Meta's UMA, trained across the periodic table and usable on a system they never saw.
- Simulations of **millions of atoms** at near-quantum accuracy.

MY OWN FIELD. THIS IS WHERE MATERIALS MODELLING CHANGED.

Generative models and inverse design

Prediction asks: given this structure, what is the property? **Inverse design asks the question a chemist actually has: given the property I want, what structure should I make?**

- **Variational autoencoders** over SELFIES: compress molecules into a continuous space, move in it, decode something new.
- **Diffusion models** for crystals. **MatterGen** (Microsoft) generates structures conditioned on a target property.
- **GNoME** (DeepMind, 2023) proposed **2.2 million** new crystals, around 380,000 of them predicted stable, against roughly 48,000 known stable inorganic crystals. How many are genuinely new and useful is disputed, which is the honest state of this subfield.

FAST-MOVING. THE BOTTLENECK IS NOW MAKING THEM.

Bayesian optimisation and active learning

You cannot run every reaction. You have a budget of perhaps twenty experiments. Which twenty?

Fit a model with error bars, then pick the experiment that is either promising or informative. Repeat.

- Reaction conditions: temperature, catalyst, solvent, time.
- **Multi-objective:** yield against cost against toxicity. There is no single best answer, only a Pareto front of honest trade-offs.
- This is the piece that turns a predictive model into an experimental plan.

Learning from imperfect data

Chemistry data is small, biased, and rarely has negative examples. Nobody publishes the reaction that failed.

- **Transfer and self-supervised learning:** pretrain on millions of unlabelled structures, fine-tune on your fifty measurements.
- **Positive-unlabelled learning:** when “not reported” does not mean “does not work”.
- **Transformers on SMILES:** the same architecture as a language model, reading molecules as sentences.

THIS IS WHAT MAKES AI USABLE IN AN ORDINARY LABORATORY, NOT JUST AT GOOGLE.

Where to look

Groups - **aimat-lab**, KIT Karlsruhe (Pascal Friederich): graph networks for materials, active learning, open datasets - **Deep Synthesis Lab**, WashU (Zhiling Zheng): the course this afternoon is built from - **Argonne**, **Berkeley Lab**, **MIT**, **DeepMind**

Things you can use today - **RDKit**: the chemistry toolkit, free - **Chemprop**: property prediction - **pymatgen** and the **Materials Project**: crystals and their data - **MACE**: pretrained potentials - **ChemMatData**: where the datasets live

HANDS-ON 1 · 50 MINUTES

Open the first notebook

WHAT YOU ARE ABOUT TO DO

Molecules in Python

1. Read a molecule from a SMILES string and draw it.
2. Turn 1128 molecules into rows of numbers.
3. Fit their solubility with **the pseudoinverse from this morning**.
4. Improve it with a fingerprint, then with a random forest.
5. See the shape of chemical space.
6. Predict a molecule you choose yourself.

EVERYTHING RUNS OFFLINE. ASK AS SOON AS SOMETHING BREAKS.

PART 2 · AFTER TEA

Agentic AI in chemistry

You have all used a chatbot.

Now watch what happens when it is allowed to act.

Model versus agent

● A language model

- Answers from what it absorbed in training
- Cannot compute, look up or measure
- Confidently wrong, and you cannot tell from the answer
- One question, one answer, done

● An agent

- A model **plus tools**, in a **loop**
- Decides for itself when to call one
- Reads the result and decides again
- Every step is visible and checkable

A tool is a function with a description

```
@tool
def molecule_properties(smiles: str) -> str:
    """Compute exact properties for a molecule
    given as a SMILES string."""
    mol = Chem.MolFromSmiles(smiles)
    return json.dumps({
        "MolWt": Descriptors.MolWt(mol),
        "LogP": Crippen.MolLogP(mol),
    })

agent = create_react_agent(llm, [molecule_properties])
```

The **docstring** is what the model reads to decide whether to call it. The code is what actually runs.

The model does not have to be right. The tools do.

Ask a language model for the molecular weight of paracetamol and it recalls a number from text it has read. Sometimes the last decimal is wrong. It cannot tell you which.

Give it RDKit and it calls RDKit. **The answer is now exact, and you can see the call that produced it.**

That is the difference between a chatbot and an instrument.

What people have built with this

- **ChemCrow** (EPFL): a language model with 18 chemistry tools. Planned and synthesised an insect repellent and three organocatalysts, on a real robotic platform.
- **Coscientist** (CMU, *Nature* 2023): read the documentation, wrote the code, and ran a palladium-catalysed coupling on a liquid handler, with very little human intervention.
- **ChemGraph** (Argonne): natural language to a real quantum-chemistry or machine-learned-potential calculation. You are about to watch it run.
- **A-Lab** (Berkeley): 17 days, 41 proposed new inorganic materials made autonomously. Chemists then challenged how many were genuinely new, and the paper was corrected. Worth knowing: the loop works, the claims need a chemist.
- **Literature mining**: agents reading thousands of papers and extracting the synthesis tables nobody had time to tabulate.

A self-driving laboratory

Planner chooses the next experiment

Executor sends it to the robot

Analyser measures the outcome

Optimiser updates the model and decides whether to continue

The result of experiment n decides experiment $n+1$, with no human in the middle.

Every block in that loop is something you have now seen: a predictive model, an acquisition rule, a set of tools, and an agent that calls them.

HANDS-ON 2 · 35 MINUTES

Open the second notebook

WHAT YOU ARE ABOUT TO DO

An agent that does chemistry

1. Make one plain call to a language model, and catch it guessing.
2. Give it three tools: exact properties, a geometry optimiser, and **the solubility model you fitted before tea**.
3. Watch the loop: it asks for a tool, reads the result, answers.
4. Ask it something of your own.

YOUR OWN FREE API KEY. NOTHING HERE COSTS MONEY.

PART 3

Where this is going

The next five years

- **Foundation models for matter.** One model pretrained across chemistry and materials, adapted to your problem with fifty measurements instead of fifty thousand.
- **Closed-loop laboratories.** Not one robot, but a design-make-test cycle that runs overnight and reports in the morning.
- **Agents as colleagues.** Reading the literature, proposing the experiment, running the calculation, writing the draft. The judgement stays yours.

None of this replaces knowing chemistry.

It removes the part of the work that was never chemistry.

IF YOU WANT TO CARRY ON

What to do next

- **Everything from today** is at github.com/Amrita-School-of-AI/ai-in-material-science. Clone it, change the dataset to something from your own laboratory.
- **CHEM 5080, AI for Experimental Chemistry** (zzhenglab.github.io/ai4chem): 25 free lectures, written for chemists with no coding background. Start at lecture 1.
- **RDKit documentation** and the **Materials Project**.
- **ChemGraph** and **scientific agent skills** on GitHub, when you want the research-grade version of this afternoon.

The reading list handout has every link.



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THANK YOU

Questions

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